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GEOGRAPHY · Winds, rainfall and climate types

UNDERSTAND IT FIRST

What is it? Wind is air moving from **high pressure to low pressure** (Day 28). Winds carry **heat and moisture** around the world, so they decide **where it rains** and **what the climate is like**. **Weather** is the condition of the air on a given day; **climate** is the **average weather over about 30 years**. India's farming depends on the **monsoon winds**, and AP faces **cyclones** almost every year, so this topic is asked often. Learn it as **forces on wind → planetary winds → periodic and local winds → jet streams and cyclones → humidity and clouds → types of rainfall → climate types (Köppen) → AP**.

1. Forces and planetary winds

- **Three forces** shape wind: the **pressure gradient force** (the bigger the pressure difference, the stronger the wind; closely packed isobars mean strong wind); the **Coriolis force** from the Earth's rotation, which turns winds to the **right in the Northern Hemisphere** and to the **left in the Southern** (**Ferrel's law**); it is **zero at the equator** and **greatest at the poles**, which is why cyclones **do not form right on the equator**; and **friction** with the ground, which slows surface winds. High up, winds blow **parallel to isobars** (**geostrophic winds**).
- **Planetary (permanent) winds:** the **trade winds** blow from the **sub-tropical highs to the equatorial low**: **north-east trades** in the north, **south-east trades** in the south; steady, they helped old sailing ships trade. The **westerlies** blow from the sub-tropical highs towards the **sub-polar lows** (30°–60°); they are strongest in the **Southern Hemisphere**, where there is little land (**'roaring forties', 'furious fifties', 'shrieking sixties'**). **Polar easterlies** blow from the polar highs. Air circulates in **three cells**: **Hadley** (tropics), **Ferrel** (mid-latitudes) and **Polar**.

Planetary winds

2. Periodic and local winds

- **Periodic winds** change with time. **Monsoons** reverse direction with the season (south-west in summer, north-east in winter over India). **Land and sea breezes:** by day the land heats faster, so a cool **sea breeze** blows onto the coast; at night a **land breeze** blows out to sea. **Mountain and valley winds:** by day warm air climbs the slopes (**valley breeze, anabatic**); at night cold air slides down (**mountain breeze, katabatic**).

Local wind	Where and what
Loo	**Hot, dry** summer wind over **north India** (May–June); can cause heatstroke.
Mango showers	Pre-monsoon showers in **Kerala and Karnataka**; help mangoes ripen. **Blossom (cherry) showers** in Karnataka help **coffee** flowers.

Kal Baisakhi (Nor'westers)	Violent evening thunderstorms in **West Bengal, Odisha and Assam** in April–May; helpful for **tea, jute and rice**.
Western disturbances	Low-pressure systems from the **Mediterranean** that bring **winter rain** to north-west India and **snow** to the Himalayas; good for **rabi wheat**.
World	**Chinook** (warm, dry, down the **Rockies**; 'snow eater'); **Foehn** (same type, **Alps**); **Sirocco** (hot, dusty, **Sahara to Italy**; 'blood rain'); **Harmattan** (dry, **West Africa**; called 'the doctor'); **Mistral** (cold, down the **Rhône valley, France**); **Bora** (cold, **Adriatic**); **Khamsin** (hot, **Egypt**); **Santa Ana** (hot, dry, **California**, spreads wildfires).

- **Jet streams** are fast winds high in the troposphere. The **sub-tropical westerly jet** lies **south of the Himalayas in winter**; when it **jumps north** in early summer, the **monsoon** can burst over India. The **tropical easterly jet** blows over the peninsula in summer.
- **Cyclones** are low-pressure systems with winds spiralling inwards: **anticlockwise in the north**, **clockwise in the south**. **Tropical cyclones** form over **warm seas (above ~27°C)**, have a calm **eye**, and hit India's east coast mostly in **May and October–December**; they are called **hurricanes** (Atlantic) and **typhoons** (west Pacific). **IMD (RSMC, New Delhi)** tracks them; names come from a list of **13 countries** (WMO/ESCAP panel). AP examples: **Hudhud (2014, Visakhapatnam)**, **Titli (2018, Srikakulam)**, **Michaung (2023, Bapatla)**. An **anticyclone** is a high-pressure system with **calm, clear** weather.

3. Humidity, clouds and rainfall

- **Humidity** is water vapour in the air. **Relative humidity** = actual vapour ÷ what the air can hold at that temperature × 100; measured with a **hygrometer** (wet-and-dry bulb). When air cools to its **dew point**, vapour **condenses** into **dew, frost, fog, mist or clouds**. **Clouds: cirrus** (high, feathery), **cumulus** (cotton heaps, fair weather), **stratus** (grey layers), **nimbus** (rain clouds); tall **cumulonimbus** give **thunder, lightning and hail**.

Three types of rainfall

Orographic rain explains India's pattern: the **windward** side of the **Western Ghats** (Konkan, Malabar) gets very heavy rain, while the **leeward rain-shadow** side (interior Karnataka, Marathwada and **Rayalaseema**) stays dry. **Mawsynram** and **Cherrapunji** on the windward Khasi hills are among the **wettest places on Earth**, while nearby **Shillong** is in the rain shadow. **Anantapur** is one of India's **driest districts**.

4. Climate types

Item	Details
Köppen's scheme	**Wladimir Köppen** grouped climates by **temperature, rainfall and vegetation** using letters: **A** tropical humid, **B** dry, **C** warm temperate, **D** cold (snow) forest, **E** polar, and **H** highland. Small letters show the season of rain: **f** (rain all year), **m** (monsoon), **w** (dry winter), **s** (dry summer).
India's types	**Amw** (Konkan, Malabar: monsoon with short dry season); **As** (**Coromandel coast** of Tamil Nadu: **dry summer, rain in winter** from the north-east monsoon); **Aw** (tropical savanna: most of the

	peninsula, including much of **AP**); **BShw** (semi-arid steppe in rain-shadow areas, e.g. parts of **Rayalaseema**); **BWhw** (hot desert: western Rajasthan); **Cwg** (Ganga plain: dry winter); **Dfc** (Arunachal); **E** (high Himalaya, Ladakh).
World types	**Equatorial** (hot, wet all year; Amazon, Congo); **tropical monsoon**; **savanna** (tall grass; Africa's game parks); **hot desert** (Sahara, Thar); **Mediterranean** (**hot dry summers, mild rainy winters**; grapes, olives, citrus); **China type**; **temperate grasslands** (**prairies, steppes, pampas, veld, downs**); **British (west European) type**; **taiga** (coniferous forest); **tundra** (mosses, lichens).
Andhra Pradesh	Most rain comes from the **south-west monsoon (June–September)**, but the **north-east monsoon (October–December)** is vital for **south coastal AP and Rayalaseema**; coastal districts face **cyclones**; summers are very hot inland.

EXAM TRAP

Pressure gradient, **Coriolis** (right in N, left in S; **zero at equator**), friction; **geostrophic** winds; **trades** (sub-tropical high → equator; NE, SE); **westerlies** (30°–60°; **roaring forties** in S); polar easterlies; **Hadley, Ferrel, Polar** cells; monsoon; **sea breeze (day), land breeze (night)**; **valley (anabatic), mountain (katabatic)**; **loo, mango showers, blossom showers, Kal Baisakhi, western disturbances**; **Chinook (Rockies), Foehn (Alps), Sirocco (Sahara–Italy), Harmattan (doctor), Mistral (Rhône), Bora, Khamsin, Santa Ana**; **sub-tropical westerly jet** and monsoon; **cyclones: anticlockwise in N**, warm seas **>27°C**, eye; hurricane, typhoon; **IMD RSMC, 13 countries**; **Hudhud 2014, Titli 2018, Michaung 2023**; relative humidity, hygrometer, dew point; **cirrus, cumulus, stratus, nimbus, cumulonimbus**; rain: **convectional, orographic (rain shadow), cyclonic**; **Mawsynram, Shillong**; Anantapur dry; **Köppen A–E, H; f, m, w, s**; India **Amw, As (Coromandel), Aw, BShw, BWhw, Cwg, Dfc, E**; **Mediterranean** (dry summer, winter rain); grasslands (**prairies, steppes, pampas, veld, downs**); AP: **SW + NE monsoon**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Coriolis force is zero at the equator.

2. In the Northern Hemisphere, winds are deflected to the right.

3. Trade winds blow from the sub-tropical high-pressure belts towards the equatorial low.

4. The westerlies are strongest in the Northern Hemisphere because of its large land masses.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The westerlies are strongest in the Southern Hemisphere, where open ocean lets them blow unchecked (roaring forties).

PRACTICE

Match List-I (local wind) with List-II (description):

List-I (local wind)	List-II (description)
A. Chinook	1. Cold wind blowing down the Rhône valley in France
B. Sirocco	2. Dry, dusty West African wind called 'the doctor'
C. Harmattan	3. Hot, dusty wind from the Sahara reaching Italy
D. Mistral	4. Warm, dry wind east of the Rockies called the 'snow eater'

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Chinook: Rockies; Sirocco: Sahara–Italy; Harmattan: West Africa; Mistral: France.

PRACTICE

Which one of the following statements is NOT correct?

A. Mango showers help mangoes ripen in Kerala and Karnataka.

B. Kal Baisakhi are evening thunderstorms in West Bengal.

C. Western disturbances bring winter rain to north-west India.

D. The loo is a cold wind that blows over Punjab in winter.

Answer: D. The loo is a hot, dry wind of the north Indian summer (May–June).

PRACTICE

Assertion (A): Shillong receives far less rain than Mawsynram, although the two are close to each other.

Reason (R): Shillong lies on the leeward (rain-shadow) side of the Khasi hills.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Moist winds drop their rain on the windward slopes; air descending on the leeward side is dry.

PRACTICE

A region has hot, dry summers and mild, rainy winters, and is famous for grapes, olives and citrus fruits. Which climate type is this?

A. Tropical monsoon

B. Mediterranean

C. Equatorial

D. Tundra

Answer: B. Winter rain with dry summers is the mark of the Mediterranean climate.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the three forces, Ferrel's law and the planetary winds with cells. 2. Write periodic winds, Indian and world local winds, jet streams and cyclones with AP examples. 3. Write humidity and clouds, the three rain types with Indian examples, and Köppen's letters with India's climate types.

ECONOMY · Economic growth and economic development

UNDERSTAND IT FIRST

What is it? Economic growth is a **rise in the output** of an economy, usually **real GDP or per capita income**: it is about **'how much'**. **Economic development** is a **broader improvement in people's lives**: less **poverty and inequality**, better **health, education, housing**, more **freedom and choices**, and a shift to better jobs: it is about **'how well'**. A country can grow fast but develop slowly if the gains go to a few. Learn it as **meanings and differences → measures of development (with India's figures) → theories of growth → Indian ideas and goals**.

1. Growth vs development

Point	Economic growth	Economic development
Nature	**Quantitative**; narrow	**Qualitative and quantitative**; broad
Measured by	Real GDP, GNP, per capita income	HDI, poverty, MPI, literacy, life expectancy, inequality, SDG Index
Time	Can happen in the **short run**	A **long-run** process; includes **structural change**
Link	**Necessary but not sufficient** for development	Usually needs growth; '**growth without development**' is possible

- **Amartya Sen** (Nobel 1998), in **'Development as Freedom' (1999)**, saw development as **expanding people's capabilities and freedoms** (to be healthy, educated, take part in society), not just income. **Sustainable development**, defined by the **Brundtland Report 'Our Common Future' (1987)**, is development that **meets present needs without harming future generations' ability** to meet theirs. **Inclusive growth** (the theme of India's **11th Five-Year Plan, 2007–12**) means growth that reaches **all sections**. **Jobless growth** is GDP rising without enough new jobs.

2. Measures of development

Measure	Details
Per capita income	The **World Bank** groups countries as **low, lower-middle, upper-middle and high income** by **GNI per capita**; India is **lower-middle income**. Simple, but hides **inequality**.
PQLI	**Physical Quality of Life Index** by **Morris D. Morris (1979)**: **life expectancy at age one, infant mortality and basic literacy**, scored 0–100.
HDI	**Human Development Index** in UNDP's **Human Development Report (since 1990)**, designed by **Mahbub ul Haq** with Amartya Sen: **health (life expectancy at birth), education (mean and expected years of schooling) and standard of living (GNI per capita, PPP)**, combined by a **geometric mean**. India's value is about **0.685** and its rank about **130 of 193** (2025 report). Related indices: **IHDI** (adjusted for inequality), **GDI**, **GII** (gender).
MPI	**Multidimensional Poverty Index** (UNDP and **Oxford's OPHI**, **2010**) looks at **health, education and living standards** together, not income alone. **NITI Aayog's national MPI** estimated that about **24.8 crore** Indians escaped multidimensional poverty between **2013–14 and 2022–23** (headcount from about **29% to 11%**).
SDGs and other measures	The UN's **Sustainable Development Goals (2015)**: **17 goals, 169 targets** for **2030**. **NITI Aayog's SDG India Index** (since 2018) scored India **71** in **2023–24**, with **Kerala and Uttarakhand** at the top. Others: **Green GDP**, **Gross National Happiness (Bhutan)**, the **World Happiness Report**.

3. Theories of growth

Rostow's stages of growth

- **Harrod–Domar model:** growth rate = **saving (investment) rate ÷ capital–output ratio**; to grow faster, **save and invest more** or use capital better. It guided India's **First Five-Year Plan (1951–56)**. **Solow (1956)** added **technology** as the main source of long-run growth.
- **Rostow's five stages (1960)** (see the diagram), with **take-off** as the key stage. **Ragnar Nurkse's 'vicious circle of poverty'**: low income → low saving → low investment → low productivity → low income. **Balanced growth** (Nurkse, **Rosenstein-Rodan's 'big push'**: invest in many sectors together) vs **unbalanced growth** (**Albert Hirschman**: invest in key sectors with strong **linkages**). **Simon Kuznets' inverted-U curve (1955)**: inequality first **rises**, then **falls** as an economy grows.
- **India: Raj Krishna** called the slow growth of about **3.5% a year (1950–80)** the **'Hindu rate of growth'**; growth sped up after the **1991 reforms**, and India has been among the **fastest-growing major economies** in recent years. National goal: **'Viksit Bharat @2047'** (a developed India by the 100th year of independence). **AP:** the **'Swarna Andhra @2047'** vision (2024) aims to make AP a **$2.4 trillion** economy with high per capita income, through ten principles such as **zero poverty, skills and water security**.

EXAM TRAP

Growth **quantitative** (real GDP, per capita); development **qualitative + quantitative** (HDI, poverty, health, education, structural change); growth **necessary, not sufficient**; **Sen 'Development as Freedom' 1999** (capabilities); **Brundtland 1987** (sustainable); **inclusive growth (11th Plan)**; jobless growth; **World Bank income groups** (India lower-middle); **PQLI Morris 1979** (life expectancy at 1, IMR, literacy); **HDI 1990, Mahbub ul Haq** (health, education, GNI PPP; geometric mean); India **~0.685, ~130/193 (2025)**; IHDI, GDI, GII; **MPI 2010 (UNDP–OPHI)**; **NITI MPI: 24.8 crore out, 29% → 11% (2013–14 to 2022–23)**; **SDGs 2015: 17 goals, 169 targets**; **SDG India Index 71 (2023–24), Kerala, Uttarakhand**; GNH; **Harrod–Domar** (s ÷ v; First Plan); **Solow** (technology); **Rostow 1960** (5 stages, take-off); **Nurkse** (vicious circle; balanced); **Rosenstein-Rodan** (big push); **Hirschman** (unbalanced, linkages); **Kuznets inverted U (1955)**; **Hindu rate ~3.5% (Raj Krishna)**; **Viksit Bharat @2047**; **Swarna Andhra @2047 ($2.4 trillion)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Economic growth is mainly a quantitative concept.

2. Economic development includes less poverty and better health and education.

3. Economic growth always leads to economic development.

4. Amartya Sen described development as the expansion of people's capabilities and freedoms.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Growth is necessary but not sufficient; if its gains are unequal, development may not follow.

PRACTICE

Match List-I (economist or model) with List-II (idea):

List-I (economist or model)	List-II (idea)
A. Harrod–Domar	1. Inequality first rises and then falls with growth
B. Rostow	2. Vicious circle of poverty
C. Nurkse	3. Stages of growth including 'take-off'
D. Kuznets	4. Growth depends on the saving rate and the capital–output ratio

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) Harrod–Domar: s/v; Rostow: stages; Nurkse: vicious circle; Kuznets: inverted U.

PRACTICE

Which one of the following statements is NOT correct?

A. The Multidimensional Poverty Index measures only income poverty.

B. The HDI uses health, education and income.

C. The PQLI was developed by Morris D. Morris.

D. The Sustainable Development Goals, adopted in 2015, have 17 goals.

Answer: A. The MPI looks at deprivations in health, education and living standards, not income alone.

PRACTICE

Assertion (A): The Harrod–Domar model was the basis of India's First Five-Year Plan.

Reason (R): The Harrod–Domar model says that technology alone, and not saving or investment,

decides the rate of growth.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: Harrod–Domar stresses saving and investment; technology-led growth is Solow's idea.

PRACTICE

A country's real GDP grows by 8% a year for a decade, but infant mortality and illiteracy hardly fall and the income gap widens. This is best described as:

A. Development without growth

B. Growth without development

C. Sustainable development

D. Balanced growth

Answer: B. Output rises but people's well-being does not improve: growth without development.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the growth vs development table and the Sen, Brundtland and inclusive-growth ideas. 2. Write the measures (per capita, PQLI, HDI, MPI, SDGs) with India's figures. 3. Write the growth theories and India's terms and goals, including Swarna Andhra.

SCIENCE · Solid waste: classification, disposal and management

UNDERSTAND IT FIRST

What is it? Solid waste is any unwanted solid material we throw away: kitchen scraps, paper, plastic, glass, old phones, hospital waste, construction debris. Indian cities produce a huge amount every day, and much of it still ends up in **open dumps** that **pollute groundwater, release methane, catch fire** and breed **mosquitoes and disease**. Good management means **segregating at source, reducing and recycling**, and safely treating the rest. As a Panchayat Secretary you deal with this directly. Learn it as **types →**

> **India's situation → disposal methods → waste hierarchy → rules for each type → conventions → Swachh Bharat and AP**.

1. Types and India's situation

- **By source: municipal solid waste** (homes, shops, markets), **industrial**, **agricultural** (crop residue), **bio-medical**, **construction and demolition (C&D)**, **e-waste**, **hazardous** (chemicals, batteries) and **plastic**. **By nature: biodegradable** (food, leaves, paper; broken down by microbes) and **non-biodegradable** (plastic, glass, metal). For collection, households separate waste into **wet** (kitchen), **dry** (paper, plastic, metal) and **domestic hazardous** (batteries, paint, expired medicines, sanitary waste).
- **India** generates about **1.6 lakh tonnes of municipal waste a day** (CPCB, 2020–21); most is collected, but only about **half is treated**, and large **legacy dumps** remain (such as **Ghazipur, Delhi**), which are now being cleared by **biomining**. About half of urban waste is **organic (wet)**, ideal for composting. India is the **world's third-largest e-waste generator**.

2. Disposal methods and the waste hierarchy

Method	How it works; merits and problems
Open dumping	Waste piled on open land: the **worst** method; causes **leachate** (toxic liquid) in groundwater, **methane**, fires, smell and disease.
Sanitary landfill	An engineered pit with a **clay or plastic liner**, **leachate collection**, daily soil cover and **methane (landfill gas) capture**; only for **inert, non-recyclable** residue.
Composting	**Aerobic** (with air) breakdown of wet waste by microbes into **manure**; **vermicomposting** uses **earthworms** (e.g. *Eisenia fetida*). Cheap; suits villages and towns.
Biomethanation	**Anaerobic** (without air) digestion of wet waste gives **biogas (mainly methane)** for fuel and **slurry** as manure (e.g. **GOBAR-Dhan** plants).
Incineration and waste-to-energy	Burning at high temperature cuts volume by about **90%** and can make **electricity**; but **mixed, wet** Indian waste burns poorly, and poor plants emit **dioxins and furans**. **Refuse-derived fuel (RDF)** from dry waste is used in **cement kilns**. Other options: **pyrolysis**, **plasma gasification**.

Waste hierarchy

The aim is a **circular economy**, where materials are kept in use instead of being thrown away. **Mission LiFE (2022)** encourages **less waste** in daily life.

3. Rules and conventions

Rule or treaty	Key points
Solid Waste Management	Made under the **Environment (Protection) Act, 1986**, replacing the 2000 rules. **Source segregation** into **wet, dry and domestic hazardous**; apply **beyond municipal areas** (census

Rules, 2016	towns, villages near cities, railways, airports, SEZs); local bodies may charge **user fees** and **spot fines**; **bulk generators** must process their own wet waste; **no open burning**; landfills only for residue. **New rules** with **four-stream segregation** (wet, dry, sanitary, special care) are replacing them (check the latest).
Plastic Waste Management Rules, 2016	Carry-bag thickness raised to **75 microns (2021)** and **120 microns (December 2022)**; **identified single-use plastic items banned from 1 July 2022**; **Extended Producer Responsibility (EPR)**: makers must collect and recycle their plastic packaging (2022 guidelines).
Other rules	**E-Waste (Management) Rules, 2022** (in force **April 2023**; EPR targets for producers, registered recyclers); **Bio-Medical Waste Management Rules, 2016** (colour-coded bags; see below); **Battery Waste Management Rules, 2022** (EPR); **Construction and Demolition Waste Rules, 2016**; **Hazardous and Other Wastes Rules, 2016**.
International conventions	**Basel (1989)**: control of **cross-border movement of hazardous waste** (plastic waste added in 2019); **Rotterdam (1998)**: **prior informed consent** for trade in hazardous chemicals and pesticides; **Stockholm (2001)**: **persistent organic pollutants (POPs)** such as DDT; **Minamata (2013)**: **mercury**. Talks on a **global plastics treaty** are continuing.

Bio-medical waste colour codes

- **Swachh Bharat Mission (Urban), 2014**; **SBM-U 2.0 (2021)** aims at **'Garbage-Free Cities'** with **star ratings** and **legacy dump clearance**; the **Swachh Survekshan** survey ranks cities (**Indore** was cleanest **seven years in a row**, 2017–2023, and now leads a 'Super Swachh League'). **SBM (Grameen) Phase II** targets **ODF Plus** villages with solid and liquid waste management. **Waste to Wealth** and **GOBAR-Dhan** turn waste into resources.
- **AP: waste-to-energy plants** near **Guntur** and **Visakhapatnam**; cities like **Vijayawada, Visakhapatnam, Tirupati and Guntur** have ranked well in Swachh Survekshan; villages use **door-to-door collection, compost sheds** (solid waste processing centres) and **green ambassadors** under the gram panchayat.

EXAM TRAP

Types: **MSW, industrial, agricultural, bio-medical, C&D, e-waste, hazardous, plastic**; biodegradable vs non-biodegradable; **wet, dry, domestic hazardous**; India **~1.6 lakh t/day (2020–21)**, **~half treated**; legacy dumps, **biomining**; **3rd largest e-waste**; **open dump (worst)**; **sanitary landfill (liner, leachate, methane)**; **composting (aerobic), vermicompost (earthworms)**; **biomethanation (anaerobic, biogas)**; **incineration (~90% volume cut; dioxins, furans)**; **RDF (cement kilns)**; hierarchy **refuse/reduce > reuse > recycle > recover > dispose**; circular economy; **Mission LiFE 2022**; **SWM Rules 2016 (EPA 1986)**: segregation, beyond municipal areas, user fees, bulk generators, **no open burning**; new **four-stream** rules; **Plastic: 75 µm (2021), 120 µm (Dec 2022), SUP ban 1 July 2022, EPR**; **E-waste Rules 2022 (Apr 2023)**; **BMW 2016**: **yellow** (anatomical, soiled), **red** (contaminated plastics), **white** (sharps), **blue** (glass, metal implants); **Battery 2022**; **C&D 2016**; **Basel 1989, Rotterdam 1998 (PIC), Stockholm 2001 (POPs), Minamata 2013 (mercury)**; **SBM-U 2014, 2.0 2021**; **Indore 7 years**; ODF Plus; AP **WtE Guntur, Visakhapatnam**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Solid Waste Management Rules, 2016:

1. They require waste to be segregated at source into wet, dry and domestic hazardous waste.

2. They apply beyond municipal areas, for example to census towns and railways.

3. They allow local bodies to collect user fees.

4. They permit open burning of waste in villages.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. The Rules prohibit open burning of waste anywhere.

PRACTICE

Match List-I (colour of bag under the Bio-Medical Waste Rules, 2016) with List-II (waste):

List-I (colour of bag under the Bio-Medical Waste Rules, 2016)	List-II (waste)
A. Yellow	**1.** Glassware and metallic implants
B. Red	**2.** Sharps such as needles
C. White (translucent)	**3.** Contaminated recyclable plastics such as tubing
D. Blue	**4.** Anatomical waste and soiled dressings

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Yellow: anatomical and soiled; red: contaminated plastics; white: sharps; blue: glass and metal implants.

PRACTICE

Which one of the following pairs is NOT correctly matched?

A. Basel Convention : cross-border movement of hazardous waste

B. Minamata Convention : mercury

C. Stockholm Convention : persistent organic pollutants

D. Rotterdam Convention : protection of the ozone layer

Answer: D. Rotterdam deals with prior informed consent for hazardous chemicals; the ozone layer is covered by the Vienna Convention and Montreal Protocol.

PRACTICE

Assertion (A): Sanitary landfills are lined at the bottom with clay or plastic liners.
Reason (R): The liners stop leachate from seeping into the groundwater.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Protecting groundwater from leachate is exactly why liners are used.

PRACTICE

A gram panchayat collects a large amount of separated kitchen and cattle waste every day and wants both cooking fuel and manure from it. Which method suits it best?

A. Incineration

B. Open dumping

C. Biomethanation (biogas plant)

D. Sanitary landfill

Answer: C. Anaerobic digestion gives biogas for cooking and slurry for manure, as in GOBAR-Dhan plants.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the types of waste and India's figures. 2. Write each disposal method with its merit and problem, and the waste hierarchy. 3. Write the rules (SWM, plastic, e-waste, bio-medical colours, battery, C&D), the conventions, Swachh Bharat and AP's examples.

NOTEBOOK PAGE · Day 29: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Winds, rainfall, climate

Area	Write
Winds	Three forces; Ferrel's law; planetary winds and cells; periodic winds; Indian and world local winds; jet streams; cyclones
Rain and climate	Humidity, condensation forms, clouds; three rain types; Köppen letters and India's types; world climate types; AP's rainfall pattern

ECONOMY: Growth and development

- Definitions and differences; Sen and Brundtland; measures (per capita, PQLI, HDI, MPI, SDG Index) with India's figures; growth theories (Harrod–Domar, Rostow, Nurkse, Kuznets, balanced vs unbalanced); Indian phrases (Hindu rate, jobless, inclusive); Viksit Bharat and Swarna Andhra.

SCIENCE: Solid waste

Write: types by source and nature; India's figures; disposal methods with merits and problems; waste hierarchy; Rules (SWM, plastic, e-waste, bio-medical colours, battery, C&D, hazardous); conventions; Swachh Bharat and AP examples.